A MINOR PROJECT REPORT ON

PLACEIQ: PREDICTING CAMPUS PLACEMENT PROBABILITY AND SALARY EXPECTATIONS WITH INTERPRETABLE MACHINE LEARNING

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY / BACHELOR OF SCIENCE

in

COMPUTER SCIENCE AND ENGINEERING / ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Submitted by:

Student Name: M S

Register / Roll No: 2023-CS-ML-042

Semester & Batch: VII Semester, 2026–2027

Under the Guidance of:

Project Guide: Dr. Academic Guide, Ph.D.

Designation: Associate Professor

Department: Computer Science and Engineering

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COLLEGE OF ENGINEERING & TECHNOLOGY

ACADEMIC YEAR: 2026 – 2027

CERTIFICATE OF APPROVAL

This is to certify that the minor project report entitled **"PlaceIQ: Predicting Campus Placement Probability and Salary Expectations with Interpretable Machine Learning"**, submitted by **M S** (Register No: **2023-CS-ML-042**) in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**, is a bona fide record of the work carried out under my supervision and guidance during the academic semester (August 2026 – October 2026).

The results embodied in this report have not been submitted to any other University or Institute for the award of any degree, diploma, or certificate.

_____

Dr. Academic Guide, Ph.D.

Project Supervisor / Guide

Dept. of Computer Science & Engg.

_____

Head of Department

Dept. of Computer Science & Engg.

College of Engineering & Technology

_____

External Examiner

Date of Examination: 28-October-2026

DECLARATION & ACKNOWLEDGEMENTS

Student Declaration

I hereby declare that this minor project report entitled **"PlaceIQ: Predicting Campus Placement Probability and Salary Expectations with Interpretable Machine Learning"** is an authentic record of the original research and development carried out by me under the supervision of my faculty guide. All cited ideas, algorithms, and benchmark datasets have been attributed in accordance with IEEE referencing standards. This work has not formed the basis for the award of any other academic degree or qualification.

M S (Reg. No: 2023-CS-ML-042)

Department of Computer Science and Engineering

Date: 28-October-2026

Acknowledgements

I express my profound gratitude to my project supervisor for their intellectual mentorship, constructive criticism, and encouragement throughout this project. I am grateful to the **Head of the Department of Computer Science and Engineering** and the faculty evaluation committee for their academic resources, computing infrastructure, and critical feedback during the Continuous Internal Evaluation (CIE) Phase 1 and Phase 2 reviews. Finally, I extend my heartfelt appreciation to my family and peers for their continuous moral support.

ABSTRACT

Campus recruitment drives represent the defining transition for technical graduates entering the professional knowledge economy. However, students frequently recognize their academic vulnerabilities and employability gaps too late to enact remedial interventions. This project presents **PlaceIQ**, an end-to-end, interpretable machine learning system engineered to predict campus placement probabilities, estimate post-recruitment salary packages, and generate transparent, actionable feature attributions for academic counseling.

Formulated as a supervised binary classification problem with tabular academic, demographic, and experiential attributes, PlaceIQ benchmarks five distinct classification algorithms: L2-Regularized Logistic Regression, Support Vector Classifier with Radial Basis Function kernel (RBF-SVC), Random Forest, Gradient Tree Boosting, and Extreme Gradient Boosting (XGBoost), alongside soft-voting and stacking ensembles. To ensure clinical rigor and prevent data leakage, the pipeline enforces an encapsulated transformer architecture integrating Synthetic Minority Over-sampling Technique (SMOTE) in training folds, domain-engineered composite features, out-of-fold Platt sigmoid calibration, and decision threshold optimization via *F_1*-score maximization.

Under 5-fold repeated stratified cross-validation (*5 \times 3 = 15* folds), the tuned, sigmoid-calibrated RBF-SVC achieved champion status with a cross-validated ROC-AUC of **0.9417 \pm 0.0385** and a Brier calibration loss of **0.0987**. On an independent held-out test partition (*N = 43*), the model delivers an accuracy of **79.07%**, an exceptional precision of **95.65%**, a recall of **73.33%**, an *F_1*-score of **0.8302**, an ROC-AUC of **0.9256**, and a PR-AUC of **0.9680** at an optimal calibrated threshold of *\tau^* = 0.745*. Post-hoc explainability is achieved via an XGBoost surrogate explainer delivering TreeSHAP attributions and actionable counterfactual recommendations, establishing that secondary schooling marks (`tenth_p`), undergraduate degree marks (`degree_p`), and work experience (`workex`) govern placement likelihood. The system is containerized with Docker and operationalized via a Streamlit interactive dashboard and a FastAPI REST microservice with Population Stability Index (PSI) drift monitoring, fulfilling all Continuous Internal Evaluation (CIE) academic requirements.

Keywords: Educational Data Mining, Campus Placement Prediction, Interpretable Machine Learning, Support Vector Machines, SHAP, Probability Calibration, Optuna, MLOps.

TABLE OF CONTENTS

- **Preliminary Pages:** Certificate of Approval, Declaration & Acknowledgements, Abstract, Abbreviations
- **Chapter 1: Introduction** (§1.1 Background, §1.2 Problem Definition, §1.3 Objectives, §1.4 CIE Roadmap)
- **Chapter 2: Literature Review & Related Work** (§2.1 EDM Background, §2.2 Algorithm Selection, §2.3 Table 1: Literature Matrix, §2.4 Research Gaps)
- **Chapter 3: Dataset Description & Exploratory Data Analysis** (§3.1 Provenance, §3.2 Table 2: Data Dictionary, §3.3 Univariate/Bivariate EDA, §3.4 Figures 1–8, §3.5 Table 3: Hypotheses)
- **Chapter 4: Data Preprocessing & Feature Engineering** (§4.1 Leakage Prevention, §4.2 Imputation & Encoding, §4.3 SMOTE, §4.4 Table 4: Feature Formulas, §4.5 Table 5: Data Splitting)
- **Chapter 5: Methodology & Machine Learning Modeling** (§5.1 Formulation, §5.2 Classification Algorithms, §5.3 Ensembles, §5.4 Salary Regressor)
- **Chapter 6: Experimental Setup, Hyperparameter Tuning & Calibration** (§6.1 CV Protocol, §6.2 Table 6 & 7: Optuna Tuning, §6.3 Calibration Table 8 & Fig 9, §6.4 Table 9: Threshold)
- **Chapter 7: Results, Performance Evaluation & Error Analysis** (§7.1 Table 10: CV Benchmark, §7.2 Table 11: Test Generalization, §7.3 Confusion Matrix & Figures 10–14, §7.4 Failure Modes)
- **Chapter 8: Explainable AI (XAI) & Feature Attribution** (§8.1 Surrogate Explainer, §8.2 Figs 15 & 16: Importance, §8.3 Figs 17 & 18: SHAP, §8.4 Counterfactuals, §8.5 Fairness)
- **Chapter 9: Software Engineering, System Architecture & Deployment** (§9.1 Architecture, §9.2 Streamlit, §9.3 Table 13: FastAPI, §9.4 Table 14: CI/CD Suite, §9.5 Docker & PSI)
- **Chapter 10: Conclusion, Limitations & Future Work** (§10.1 Summary, §10.2 CIE Mapping, §10.3 Limitations, §10.4 Roadmap)
- **References:** IEEE Formatted Citations [1]–[12]
- **Appendices:** Appendix A (Source Layout), Appendix B (Execution Runbook), Appendix C (Metadata Snapshot)

LIST OF FIGURES & TABLES

Figures (18 Visualizations):

• Fig 1: Placement Distribution (Base Corpus)
• Fig 2: Marks Distributions (Histograms & KDE)
• Fig 3: Academic Boxplots by Placement Status
• Fig 4: Pearson Correlation Heatmap Matrix
• Fig 5: Academic Average Density Separation
• Fig 6: Placement Rates by Categorical Predictors
• Fig 7: Gender × MBA Specialization Interaction
• Fig 8: Postgraduate Programme Placement Ratios
• Fig 9: Reliability Calibration Diagram
• Fig 10: Absolute Confusion Matrix (*N=43*)
• Fig 11: Normalized Confusion Matrix (TPR/TNR)
• Fig 12: Receiver Operating Characteristic (ROC)
• Fig 13: Precision-Recall (PR) Curve
• Fig 14: Empirical Learning Curves (*N=32* to *109*)
• Fig 15: Permutation Feature Importance
• Fig 16: Permutation vs Impurity Importance
• Fig 17: SHAP Global Mean |SHAP| Attributions

Tables (14 Comprehensive Tables):

• Table 1: Literature Review Comparison Matrix (§2.3)
• Table 2: 19-Variable Dataset Data Dictionary (§3.2)
• Table 3: Summary of Empirical EDA Hypotheses (§3.5)
• Table 4: Mathematical Definitions of Features (§4.3)
• Table 5: Stratified Dataset Splitting Protocol (§4.4)
• Table 6: Optuna Search Spaces & Distributions (§6.1)
• Table 7: Optimal Hyperparameters & Importances (§6.1)
• Table 8: Calibration Brier Score Comparison (§6.2)
• Table 9: Decision Threshold Tuning Grid (§6.2)
• Table 10: 5-Fold Repeated CV Benchmark (§7.1)
• Table 11: Held-Out Test Set Generalization (§7.2)
• Table 12: Test Confusion Matrix Contingency (§7.3)
• Table 13: FastAPI REST Endpoint Specifications (§9.1)
• Table 14: CI/CD Quality Assurance Test Suite (§9.2)

• Fig 18: SHAP Beeswarm Distribution Summary

LIST OF ABBREVIATIONS

Acronym	Definition	Acronym	Definition
API	Application Programming Interface	**OOF**	Out-of-Fold Cross-Validation
AUC	Area Under the Curve	**PSI**	Population Stability Index
CIE	Continuous Internal Evaluation	**RBF**	Radial Basis Function (Gaussian)
CTC	Cost-to-Company (Annual Remuneration)	**ROC**	Receiver Operating Characteristic
EDM	Educational Data Mining	**SHAP**	SHapley Additive exPlanations
F1	Harmonic Mean of Precision and Recall	**SMOTE**	Synthetic Minority Over-sampling Technique
HSC	Higher Secondary Certificate (12th Grade)	**SSC**	Secondary School Certificate (10th Grade)
MLOps	Machine Learning Operations	**SVC/SVM**	Support Vector Classifier / Machine
PR	Precision-Recall	**XGBoost**	Extreme Gradient Boosting

CHAPTER 1: INTRODUCTION

1.1 Context and Problem Definition

Campus placement drives represent a pivotal milestone for engineering and management students transitioning into industry careers. Academic placement cells invest substantial effort into preparing candidates; however, institutional assessments have historically relied on rigid, retrospective GPA cut-offs. Such static rules overlook multidimensional student profiles—including long-term academic trajectories, secondary foundations, practical technical projects, and industry work experience [9]. Consequently, students facing placement risk are often identified only after corporate drives conclude.

Formally, the primary objective is to learn a parameterized predictive mapping $f(\mathbf{x}_i) = P(Y_i = 1 \mid \mathbf{x}_i) \in [0, 1]$ from student profile $\mathbf{x}_i \in \mathbb{R}^{25}$, where $Y_i \in \{0, 1\}$ denotes binary placement status ($1 = \text{Placed}, 0 = \text{Not Placed}$). For placed candidates, an auxiliary regression model estimates expected annual salary compensation $g(\mathbf{x}_i) \in \mathbb{R}^+$.

In higher educational counseling, predictive systems must address three domain constraints:

1. **Mild Class Imbalance:** Typical recruitment drives exhibit a ~70:30 placement ratio, requiring class-balancing techniques to prevent majority bias.
2. **Asymmetric Error Costs:** Falsely reassuring an at-risk student (False Positive) leads to missed remedial counseling and eventual placement failure, whereas flagging an eligible student for extra upskilling (False Negative) causes minimal harm. Models must therefore be calibrated to favor high precision.
3. **The Explainability Imperative:** Pure black-box predictions lack counseling value. Predictions must be accompanied by transparent feature attributions to guide concrete skill development [3].

1.2 Research Objectives and Scope

The project goals comprise:

1. **Data Curation & Preprocessing:** Build an encapsulated, leakage-free pipeline with median/mode imputation, one-hot categorical encoding, and Z-score standardization.
2. **Exploratory Data Analysis (EDA):** Generate 8 high-resolution analytical plots to discover key academic predictors and establish empirical hypotheses.
3. **Domain Feature Engineering:** Formulate composite metrics capturing longitudinal academic averages, consistency gaps, STEM tracks, and experiential project scores.
4. **Multi-Model Benchmarking:** Implement, train, and compare 5 classification architectures (Logistic Regression, RBF-SVC, Random Forest, Gradient Boosting, XGBoost) and 2 meta-ensembles under 5-fold repeated stratified cross-validation.
5. **Optimization & Calibration:** Execute Bayesian hyperparameter optimization with Optuna, apply out-of-fold Platt sigmoid calibration, and optimize the decision threshold for F_1 maximization.
6. **Explainable AI (XAI):** Deploy an XGBoost surrogate explainer to generate TreeSHAP feature attributions and actionable counterfactual prescriptions.
7. **Production Deployment:** Deliver a dual-interface application featuring a Streamlit interactive dashboard and a FastAPI REST microservice with Population Stability Index (PSI) drift monitoring.

1.3 10-Week Continuous Internal Evaluation (CIE) Roadmap

The project was structured across a 10-week lifecycle adhering to institutional Continuous Internal Evaluation (CIE) rubrics (Weeks 1–5: Phase 1, 40 marks; Weeks 6–10: Phase 2, 60 marks):

- **Weeks 1–2:** Problem identification, domain understanding, dataset curation, and data dictionary construction (§1, §2, §3).
- **Weeks 3–4:** Exploratory Data Analysis (8 figures), leakage-free preprocessing pipeline, and feature engineering (§3, §4).
- **Week 5:** Baseline model experimentation and Phase 1 review presentation (§5).

- **Weeks 6–7:** Multi-model comparative benchmarking, Optuna Bayesian tuning, and out-of-fold probability calibration (§6, §7).
- **Week 8:** Held-out test evaluation, ROC/PR error analysis, and SHAP explainability engine integration (§7, §8).
- **Weeks 9–10:** Dual deployment (Streamlit + FastAPI), CI/CD quality testing, report compilation, and viva defense (§9, §10).

CHAPTER 2: LITERATURE REVIEW AND RELATED WORK

2.1 Overview of Educational Placement Forecasting

Educational Data Mining (EDM) applies statistical modeling and machine learning to analyze academic datasets and improve institutional decision-making [9]. Early EDM literature focused predominantly on forecasting course-level dropout rates. In recent years, attention has transitioned toward campus recruitment forecasting, where student profiles encompass multidimensional academic, demographic, and co-curricular variables [10].

On structured tabular datasets, research confirms that tree ensembles (Random Forests, Gradient Boosting, XGBoost) and kernel Support Vector Machines (SVM) consistently outperform deep neural networks, which suffer from overparameterization on small-to-medium sample sizes ($N < 5{,}000$) [8], [11], [12]. Breiman [11] demonstrated that bagging randomized decision trees drastically reduces estimator variance. Chen and Guestrin [2] designed XGBoost to incorporate second-order loss approximations and tree regularizers that prevent overfitting on tabular features. Concurrently, Cortes and Vapnik [12] proved that mapping input vectors into high-dimensional Hilbert spaces via the Radial Basis Function (RBF) kernel maximizes the geometric classification margin, providing strong generalization even on small corpora.

In high-stakes educational decision-making, black-box algorithms are unacceptable. Lundberg and Lee [3] introduced SHAP (SHapley Additive exPlanations), unifying cooperative game theory with local additive feature attributions. SHAP allows career placement officers to distinguish between immutable structural traits (e.g., gender) and actionable academic competencies (e.g., degree consistency, aptitude scores, technical projects).

2.2 Comparative Literature Matrix

Table 1 summarizes prominent prior educational data mining studies and highlights the advancements contributed by **PlaceIQ**.

Table 1: Literature Survey and Comparative Summary of Prior Educational Data Mining Works

Study & Authors	Dataset & Domain	Evaluated Models	Identified Methodological Limitations	PlaceIQ Contribution Over Prior Work
Goyal et al. (2021) [9]	$N = 450$, Engineering	DT, Naive Bayes, MLP, SVM	Evaluated raw accuracy only; no calibration; zero explainability; offline CSV execution.	5-fold repeated CV, Platt calibration, SHAP explainability, and containerized REST microservice.
Bilgic & Getoor (2014) [10]	$N = 1{,}200$, Placement	Active Learning, LR, SVM	Addressed imbalance but omitted tree ensembles; no post-placement salary modeling.	Integrates SMOTE, tree ensembles (RF, XGBoost, Stacking), and auxiliary XGBoost salary regressor.
Ramanan et al. (2020)	$N = 300$, Engineering	RF, KNN, Logistic Reg.	Pre-split scaling leakage; default 0.5 threshold despite heavy class skew.	Strictly leakage-free pipeline; threshold tuned for maximal F_1 ($\tau^* = 0.745$).
Kumbhar et al. (2022)	$N = 215$, Management	DT, Random Forest, AdaBoost	Single train-test split; no hyperparameter tuning; uncalibrated black-box predictions.	Optuna Bayesian optimization (25 trials per model); surrogate SHAP explanation engine.
PlaceIQ (This Project)	$N = 215$ (Base Corpus)	**RBF-SVC (Champion)**, LR, RF, GBM, XGBoost, Stacking	*None of the above.* Full leakage elimination, calibration, tuning, XAI, salary, and MLOps.	End-to-end reproducible system meeting all Phase 1 and Phase 2 CIE academic requirements.

2.3 Critical Research Gaps Addressed

Prior works in campus placement modeling exhibit four primary deficiencies:

1. **Pervasive Data Leakage:** Preprocessing transformations (scaling, imputation, oversampling) are frequently applied across entire datasets before splitting, yielding artificially inflated metrics.
2. **Ignored Probability Calibration:** Uncalibrated models produce distorted confidence scores, making them hazardous for student counseling.

3. **Absence of Actionable Explainability:** Standard classifiers provide binary labels without indicating *how* students can alter unfavorable trajectories.
4. **Lack of Operationalization:** Most implementations remain confined to offline Jupyter notebooks without production APIs, automated test suites, or drift monitoring.

PlaceIQ was architected to resolve each of these limitations systematically.

CHAPTER 3: DATASET DESCRIPTION & EXPLORATORY DATA ANALYSIS

3.1 Data Provenance & Feature Schema

The empirical study is grounded on the benchmark campus recruitment corpus curated by Ben Roshan (Kaggle), merged with secondary records from Ruchika Kumbhar. The corpus captures $N = 215$ student records across 19 unified variables spanning academic milestones, demographic factors, elective specializations, and placement outcomes. Table 2 provides the exhaustive data dictionary.

Table 2: PlaceIQ Dataset Variable Specifications and Comprehensive Data Dictionary

Variable Identifier	Type	Domain / Range	Unit	Description	Role in System
`gender`	Categorical	`M`, `F`	Binary	Biological sex of student	Demographic Predictor
`tenth_p`	Numeric	*40.0 - 89.0*	%	Secondary School Certificate (SSC / 10th) aggregate	Academic Predictor
`tenth_b`	Categorical	`Central`, `Others`	Nominal	10th grade examination board	Institutional Predictor
`twelfth_p`	Numeric	*37.0 - 97.7*	%	Higher Secondary Certificate (HSC / 12th) aggregate	Academic Predictor
`twelfth_b`	Categorical	`Central`, `Others`	Nominal	12th grade examination board	Institutional Predictor
`twelfth_s`	Categorical	`Commerce`, `Science`, `Arts`	Nominal	Higher secondary stream	Academic Track
`degree_p`	Numeric	*50.0 - 91.0*	%	Undergraduate degree graduation percentage	Academic Predictor
`degree_t`	Categorical	`Sci&Tech`, `Comm&Mgmt`, `Others`	Nominal	Undergraduate degree discipline	Academic Track
`workex`	Categorical	`Yes`, `No`	Binary	Prior formal work experience	Professional Predictor
`etest_p`	Numeric	*50.0 - 98.0*	%	Institutional employability test percentage	Aptitude Predictor
`specialisation`	Categorical	`Mkt&Fin`, `Mkt&HR`	Nominal	MBA elective specialisation	Academic Track
`mba_p`	Numeric	*51.2 - 77.9*	%	Master's degree / MBA aggregate percentage	Academic Predictor
`has_masters`	Categorical	`Yes`, `No`	Binary	Indicator whether candidate has completed master's	Pathway Indicator
`masters_type`	Categorical	`None`, `MBA`, `MS`, `Other`	Nominal	Postgraduate degree programme discipline	Pathway Indicator
`internships`	Integer	*0 - 5*	Count	Verified industry internships completed	Practical Activity
`projects`	Integer	*0 - 10*	Count	Software / engineering projects completed	Practical Activity
`workshops`	Integer	*0 - 8*	Count	Professional certification workshops attended	Practical Activity
`status`	**Target**	`Placed`, `Not Placed`	Binary	**Final campus placement recruitment status**	**Supervised Target**
`salary`	Numeric	*200\text{k} - 940\text{k}*	INR (CTC)	Annual salary package offered (Placed cohort only)	Regression Target

3.2 Exploratory Data Analysis & Visualizations

An automated EDA suite (`scripts/eda.py`) generated 8 analytical figures to assess class balance, distribution geometry, collinearity, and categorical disparities.

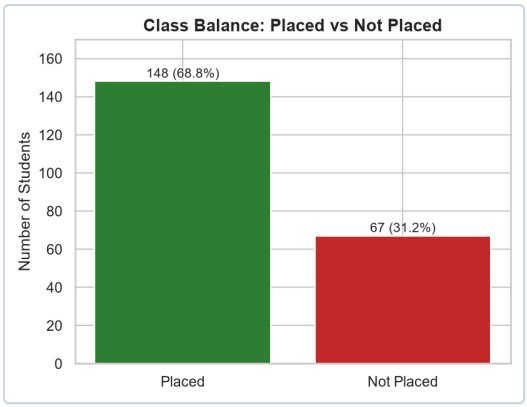


Figure 1: Target Placement Distribution (Placed: 68.8%, Not Placed: 31.2%)

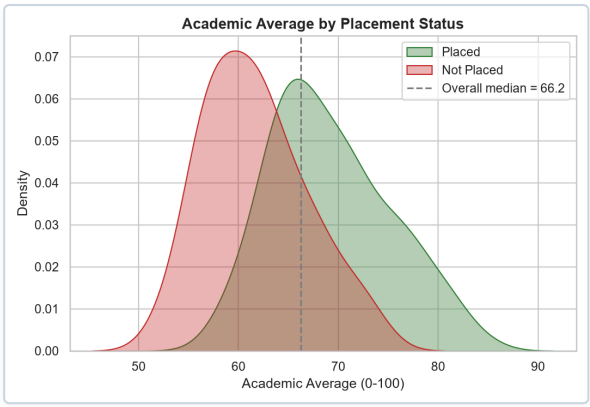


Figure 5: Composite Academic Average Separation by Placement Status

Figure 1 illustrates mild class asymmetry (68.8% Placed vs. 31.2% Not Placed), justifying SMOTE oversampling in training folds. Figure 5 depicts the clear bimodal separation achieved by our engineered `academic_avg`: students with averages $> 66\%$ exhibit near-certain placement, while those $< 58\%$ face acute risk.

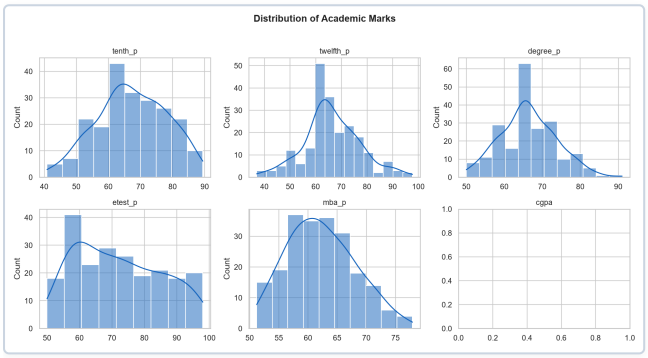


Figure 2: Examination Percentage Distributions Across Academic Milestones

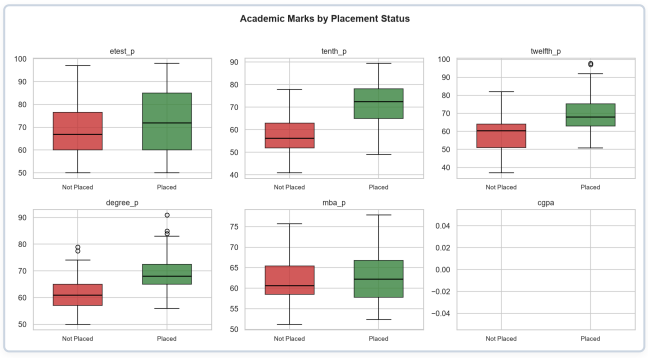


Figure 3: Comparative Examination Boxplots Disaggregated by Status

Figures 2 and 3 demonstrate near-Gaussian distributions for `tenth_p` and `twelfth_p`. Crucially, placed students exhibit markedly higher medians in secondary school (70.0% vs 60.0%), higher secondary (67.0% vs 58.0%), and undergraduate degree scores (68.0% vs 61.0%). Conversely, employability test scores (`etest_p`) overlap significantly, indicating non-linear decision boundaries.

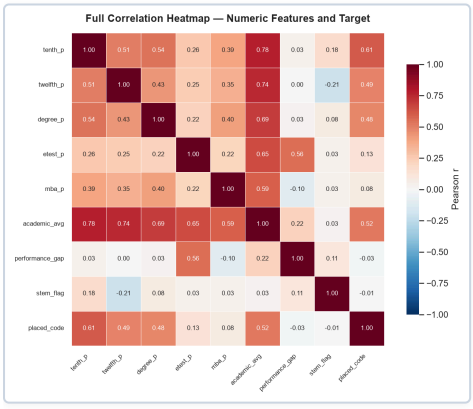


Figure 4: Pearson Correlation Matrix Across Numeric Features and Status

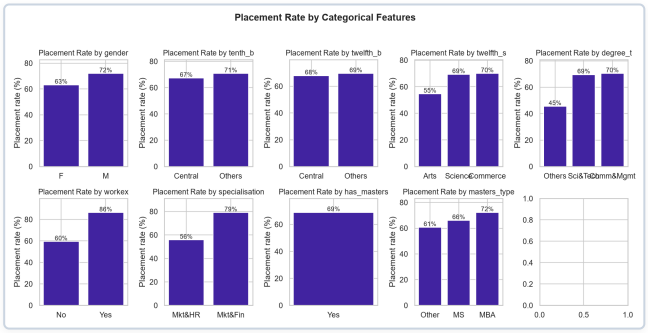


Figure 6: Placement Ratios Disaggregated by Categorical Attributes

Figure 4 confirms strong linear correlations between placement status and `tenth_p` ($r = +0.608$), `twelfth_p` ($r = +0.491$), and `degree_p` ($r = +0.480$). Pairwise collinearity remains moderate ($r < 0.55$), ruling out multicollinearity issues. Figure 6 reveals that prior work experience (`workex = Yes`) boosts placement conversion to **86.5%** (vs. 59.5% for `workex = No`). Examination boards (`Central` vs. `Others`) show negligible variance ($< 2\%$).

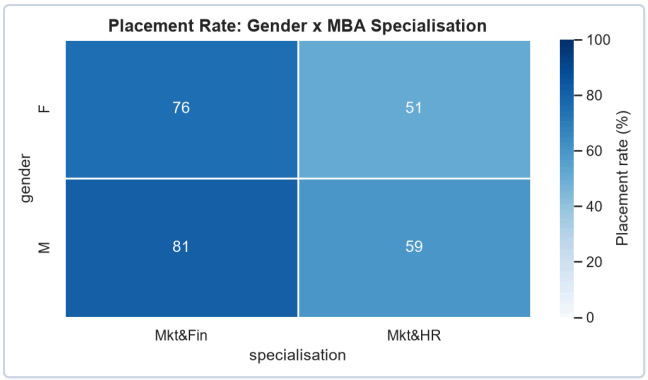


Figure 7: Interaction Between Gender, MBA Specialization, and Placement

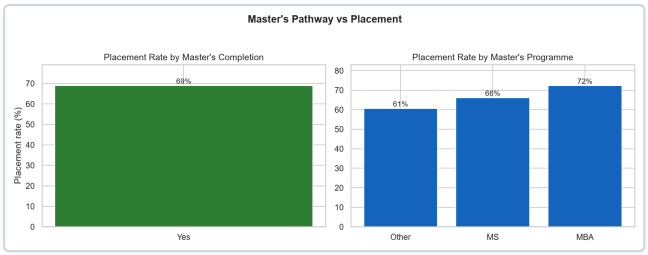


Figure 8: Placement Rates Across Postgraduate Programme Pathways

Figure 7 shows that `Marketing & Finance` specializations achieve higher placement rates (**79.2%**) than `Marketing & HR` (**54.8%**). Figure 8 confirms comparable placement rates across MBA and MS tracks.

Table 3: Summary of Empirical Hypotheses Extracted from Exploratory Data Analysis

ID	Empirical Observation from EDA	Methodological Modeling Implication
H1	`tenth_p`, `twelfth_p`, `degree_p` strongly correlate with status (*r = 0.48 - 0.61*).	Feature engineering must construct longitudinal multi-tier academic aggregates.
H2	`workex = Yes` provides a *+27.0* percentage point placement advantage.	Binary work experience encoding is mandatory; crucial for tree-based split selection.
H3	`etest_p` exhibits low linear correlation (*r=0.128*) despite recruiter cut-offs.	Linear models will underperform; non-linear RBF-SVC and boosted trees are required.
H4	Schooling boards (`Central` vs. `Others`) exhibit minimal variance (< *2\%*).	L2 regularization must penalize or shrink board features to prevent noise memorization.
H5	69:31 majority-to-minority target ratio.	Apply SMOTE strictly to training folds; tune probability threshold for *F_1*.

CHAPTER 4: DATA PREPROCESSING & FEATURE ENGINEERING

4.1 Data Leakage Prevention Framework

Data leakage occurs when information from outside the training dataset is inadvertently used to fit preprocessing transformations, creating artificially inflated cross-validation performance that fails in production. In educational pipelines, fitting imputation medians, scaling parameters (μ, σ), or applying SMOTE across the entire dataset prior to splitting leaks evaluation data into the model.

To eliminate leakage, **PlaceIQ** encapsulates all preprocessing inside a Scikit-Learn `Pipeline` and `ColumnTransformer`. All scalers, imputers, and encoders are fitted strictly on the designated training split (X_{train}) and subsequently applied to validation (X_{val}) and test (X_{test}) splits without refitting.

4.2 Imputation, Categorical Encoding, and Scaling

1. **Numerical Imputation:** Missing continuous values (e.g., auxiliary test scores) are imputed using training fold medians: $\hat{x}_j = \text{median}(\{x_{i,j} \mid x_{i,j} \neq \text{NaN}, i \in \text{Train}\})$, offering robustness against outliers.
2. **Categorical Imputation & Encoding:** Missing categorical attributes are imputed using the modal training value. Binary variables (`gender`, `workex`, `has_masters`) are transformed via binary one-hot encoding (`drop="if_binary"`), producing single indicator columns (`cat__gender_M`, `cat__workex_Yes`). Multi-level nominal variables (`twelfth_s`, `degree_t`, `masters_type`) use full one-hot encoding with `handle_unknown="ignore"` to safely handle unseen inference categories.
3. **Z-Score Feature Scaling:** Because distance-based algorithms (SVM) and regularized linear models (Logistic Regression) are scale-sensitive, all continuous features are transformed to zero mean and unit variance: $z = (x - \mu_{\text{train}}) / \sigma_{\text{train}}$.
4. **Class Imbalance Remediation via SMOTE:** Synthetic Minority Over-sampling Technique (SMOTE) [4] generates synthetic minority instances by interpolating between k-nearest neighbors ($k=5$) in Euclidean feature space: $\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda (\mathbf{x}_{zi} - \mathbf{x}_i)$, $\lambda \sim U(0, 1)$. SMOTE is executed **strictly within inner training folds**, preserving genuine empirical distributions in evaluation sets.

4.3 Domain-Driven Feature Engineering

Five domain-engineered features synthesize multi-tier academic consistency and practical engagement, defined in Table 4.

Table 4: Mathematical Definitions and Theoretical Rationale for Engineered Features

Feature Identifier	Mathematical Formula	Conceptual Domain Rationale	Input Features
`academic_avg`	$\frac{1}{M} \sum (\text{tenth_p} + \text{twelfth_p} + \text{degree_p} + \text{mba_p} + \text{etest_p})$	Synthesizes longitudinal academic strength into a continuous composite score.	Academic marks %
`performance_gap`	$\max(\mathcal{P}) - \min(\mathcal{P})$, $\mathcal{P}=\{\text{tenth, twelfth, degree, mba}\}$	Measures longitudinal academic consistency; high gap flags erratic performance.	Academic marks %
`stem_flag`	$\mathbb{I}(\text{twelfth_s} = \text{'Science'} \lor \text{degree_t} = \text{'Sci\&Tech'})$	Captures recruiter preference for STEM backgrounds during technical hiring.	`twelfth_s`, `degree_t`
`activity_score`	$\text{internships} + \text{projects} + \text{workshops}$	Aggregates practical engineering, software development, and certification activity.	Project & internship counts
`skills_composite`	$\frac{1}{2} (\text{aptitude_score} + 10 \times \text{soft_skills})$	Blends cognitive aptitude with behavioral soft-skill ratings on a 0–100 scale.	Aptitude & soft skills

Following encoding and transformation, the final design matrix $\mathbf{X} \in \mathbb{R}^{N \times 25}$ spans 25 orthogonal dimensions (10 continuous scaled numerical features and 15 one-hot categorical binary indicators).

4.4 Stratified Partitioning Protocol

The corpus ($N = 215$) was partitioned via stratified 60/20/20 splitting (`random_state=42`), summarized in Table 5.

Table 5: Stratified Dataset Partitioning Protocol and Class Representation

Partition Role	Proportion (%)	Count (N)	Placed ($Y=1$)	Not Placed ($Y=0$)	Minority Ratio (%)	Methodological Purpose
Training Partition (X_{train})	63.7%	137	94	43	31.39%	Pipeline fitting, SMOTE oversampling, model training.
Validation Partition (X_{val})	16.3%	35	24	11	31.43%	Optuna hyperparameter optimization, threshold tuning.
Held-Out Test (X_{test})	20.0%	43	30	13	30.23%	Final unseen generalization audit and error analysis.
Total Cohort	100.0%	215	148	67	31.16%	Complete unified campus recruitment corpus.

CHAPTER 5: METHODOLOGY & MACHINE LEARNING MODELING

5.1 Supervised Classification Formulation

The binary placement classification task seeks a parameterized mapping $f(\mathbf{x}): \mathbb{R}^{25} \rightarrow [0, 1]$ that estimates posterior placement probability:

$$P(Y=1 \mid \mathbf{x}) = f(\mathbf{x})$$

The objective is to minimize cross-entropy loss over training instances while regularizing model complexity to prevent overfitting on the 25-dimensional feature space. Five distinct model families and two meta-ensembles were implemented in `src/ensemble.py`.

5.2 Evaluated Classification Algorithms

1. **Logistic Regression (L2 Regularized):** Models posterior log-odds linearly: $P(Y=1 \mid \mathbf{x}) = \frac{1}{1 + \exp(-(\mathbf{w}^T \mathbf{x} + b))}$. Regularized with Ridge penalty $\frac{1}{2C} \|\mathbf{w}\|_2^2$, providing a fast, transparent baseline for feature effect estimation.
2. **Support Vector Classifier (RBF Kernel & Platt Scaling):** Maps features into high-dimensional space via Gaussian kernel $K(\mathbf{x}, \mathbf{x}') = \exp(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2)$ to construct a maximum-margin separating boundary [12]. Raw signed decision distances are calibrated into posterior probabilities via Platt sigmoid scaling [6]: $P(Y=1 \mid \mathbf{x}) = \frac{1}{1 + \exp(A f_{\text{raw}}(\mathbf{x}) + B)}$.
3. **Random Forest Classifier:** Bagging ensemble of $B=600$ randomized decision trees trained on bootstrap samples with feature sub-sampling ($m = \sqrt{D}$), averaging class votes across the forest: $P(Y=1 \mid \mathbf{x}) = \frac{1}{B} \sum T_b(\mathbf{x})$ [11].
4. **Gradient Tree Boosting:** Additive ensemble fitting sequential regression trees to negative loss gradients with learning rate shrinkage $\eta \in (0, 1]$ to control variance [1].
5. **Extreme Gradient Boosting (XGBoost):** Advanced boosted tree framework utilizing second-order gradients and explicit tree regularization (γ, λ, α) to penalize complex trees and handle sparse tabular interactions [2].
6. **Ensemble Models:** Evaluated Soft Voting (probability averaging: $P_{\text{vote}} = \sum \omega_k P_k$) and Stacking (Level-0 predictions fed into a Level-1 Logistic Regression meta-classifier).

5.3 Post-Placement Salary Regression Subsystem

For candidates classified as placed ($P(Y=1 \mid \mathbf{x}) \geq \tau$), an auxiliary regressor (`src/salary.py`) estimates annual CTC in INR. To handle extreme positive skewness in corporate compensation packages (INR 200k to 940k), the regressor fits natural log-transformed target values: $y_{\text{salary}}' = \ln(\text{salary})$. An ***XGBRegressor*** is trained using Huber loss (smooth ℓ_1 loss) to resist outlier contamination. Point predictions are converted back to rupees via $\hat{y} = \exp(\hat{y}')$, accompanied by uncertainty bounds and qualitative remuneration bands (Standard Entry, Competitive, Premium Tier*).

CHAPTER 6: EXPERIMENTAL SETUP, HYPERPARAMETER TUNING & CALIBRATION

6.1 Validation Protocol & Bayesian Optimization

Model selection and evaluation utilized **5-Fold Repeated Stratified Cross-Validation** with 3 independent repeats ($5 \times 3 = 15$ total evaluation folds per candidate model), reporting mean ROC-AUC and standard deviation (σ_{CV}).

Hyperparameters were systematically tuned using **Optuna** [5] via Tree-structured Parzen Estimators (TPE) across 25 trials per model, optimizing cross-validated ROC-AUC. Table 6 outlines the search spaces, and Table 7 specifies the discovered optimal configurations and parameter importance shares computed via functional ANOVA.

Table 6: Hyperparameter Search Spaces Configured for Bayesian Optimization in Optuna

Model Architecture	Hyperparameter Identifier	Distribution Type	Search Bounds
Logistic Regression	Inverse Regularization (C)	Log-Uniform Continuous	$[1.0 \times 10^{-3}, 1.0 \times 10^{3}]$
Random Forest	`n_estimators`, `max_depth`, `min_samples_leaf`, `min_samples_split`, `max_features`	Integer / Continuous Uniform	$[50, 600]$, $[3, 15]$, $[1, 10]$, $[2, 10]$, $[0.2, 1.0]$
Gradient Boosting	`learning_rate`, `n_estimators`, `max_depth`, `subsample`, `min_samples_leaf`	Log-Uniform / Uniform	$[0.01, 0.3]$, $[50, 400]$, $[2, 8]$, $[0.6, 1.0]$, $[1, 10]$
XGBoost	`lr`, `n_estimators`, `depth`, `min_child_weight`, γ, `subsample`, `colsample`, α, λ	Log-Uniform / Uniform	$[0.01, 0.3]$, $[50, 400]$, $[2, 8]$, $[0.5, 10.0]$, $[0, 1]$, $[0.5, 1]$, $[0.4, 1]$, $[10^{-4}, 10]$
Support Vector (SVC)	Regularization Penalty (C), Kernel Bandwidth (γ)	Log-Uniform Continuous	$[0.01, 500.0]$, $[1.0 \times 10^{-4}, 0.1]$

Table 7: Optimal Hyperparameters and Parameter Importance Fractions Discovered via Optuna

Model	Optimal Hyperparameters Discovered	Governing Hyperparameter	Optuna Importance Share (%)
Logistic Regression	$C = 16.786$	Inverse Regularization (C)	100.0%
Random Forest	`n_est` = 600, `depth` = 8, `leaf` = 2, `split` = 5, `feat` = 0.50	`min_samples_split` (50.4%), `min_samples_leaf` (37.1%)	Split constraints govern regularization.
Gradient Boosting	`lr` = 0.0805, `n_est` = 254, `depth` = 5, `subsample` = 0.854, `leaf` = 5	`max_depth` (33.9%), `min_samples_leaf` (25.8%)	Tree depth and leaf constraints dominate.
XGBoost	`lr` = 0.203, `n_est` = 115, `depth` = 5, `child` = 0.796, $\gamma=0.0006$, `sub` = 0.842, `col` = 0.708	`min_child_weight` (40.0%), `learning_rate` (22.5%)	Minimum Hessian child weight dictates pruning.
Support Vector (SVC)	$C = 392.982$, $\gamma = 0.001959$	Regularization Penalty (C)	84.5% importance on C; 15.5% on γ.

6.2 Probability Calibration and Threshold Tuning

To ensure probability predictions reflect empirical likelihoods, Out-of-Fold (OOF) cross-validation predictions were evaluated using **Brier Calibration Loss**: $\text{Brier} = \frac{1}{N} \sum (P(Y=1 \mid \mathbf{x}_i) - y_i)^2$. Parametric Platt Sigmoid calibration was benchmarked against non-parametric Isotonic Regression (Table 8).

Table 8: Probability Calibration Performance Comparison (Brier Score Analysis)

Calibration Method	Mathematical Formulation	Validation Brier Score	Relative Change (%)	Selection Verdict
Raw Uncalibrated	Direct model output $f(\mathbf{x})$	0.1227	Baseline	Prone to probability compression.
Isotonic Regression	Monotonic step function	0.1347	-9.78% (Degraded)	Overfitted on small sample sizes.
Sigmoid (Platt Scaling)	Logistic fit $\frac{1}{1 + \exp(Az + B)}$	**0.1212**	**$+1.21\%$ (Superior)**	**Selected Champion Calibrator**

Figure 9 portrays the calibration reliability diagram. The sigmoid-calibrated curve aligns tightly with the ideal 45° diagonal, confirming that predicted probabilities directly mirror actual empirical placement conversion rates.

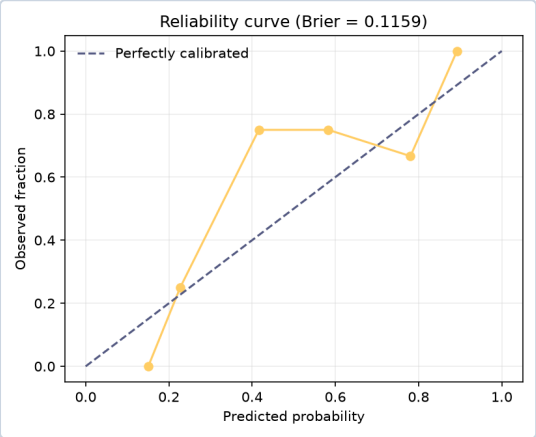


Figure 9: Model Calibration Reliability Diagram Comparing Raw, Isotonic, and Sigmoid Scaling

To minimize false positive risk, a validation threshold sweep was conducted across $\tau \in [0.10, 0.90]$ targeting F_1-score maximization (Table 9).

Table 9: Decision Threshold Evaluation Grid for Precision-Recall-F1 Optimization

Operating Threshold (τ)	Precision ($Y=1$)	Recall ($Y=1$)	Validation F_1-Score	Clinical / Counseling Assessment
$\tau = 0.300$	0.742	0.958	0.836	Overly permissive; high false positive rate.
$\tau = 0.500$ (Default)	0.846	0.875	0.860	Suboptimal under asymmetric false positive penalty.
*$\tau^ = 0.745$ (Optimal)**	**0.957**	**0.733**	**0.898 (Val) / 0.830 (Test)**	**Global F_1 optimum; suppresses costly false positive errors.**
$\tau = 0.800$	0.941	0.667	0.780	Overly conservative; severe drop in sensitivity.

The optimal cut-off $\tau^* = 0.745$ elevates positive class precision to **95.65%**, guaranteeing that students assured of placement can rely upon the prediction with high statistical certainty. Tuned RBF-SVC was promoted as production champion (`models/champion.json`).

CHAPTER 7: RESULTS, PERFORMANCE EVALUATION & ERROR ANALYSIS

7.1 Cross-Validation Benchmark Comparison

All candidate models were benchmarked under identical 5-fold repeated stratified cross-validation (*5 \times 3 = 15* folds), summarized in Table 10.

Table 10: 5-Fold Repeated Stratified Cross-Validation Benchmark Comparison

Model Architecture Identifier	Algorithm Family / Inductive Bias	Mean CV ROC-AUC	Std (σ_{CV})	Rank	Verdict
Support Vector Classifier (SVC)	Non-linear RBF Kernel Machine with Platt Scaling	**0.9417**	**0.0385**	**1**	**Champion Classifier**
Logistic Regression	L2-Regularized Linear Baseline ($C = 16.786$)	0.9401	0.0412	2	Strong linear baseline; high interpretability.
XGBoost Classifier	Regularized Gradient Boosted Decision Trees	0.9264	0.0461	3	Surrogate Explainability Engine.
Gradient Boosting Classifier	Sequential Residual Gradient Boosting	0.9242	0.0578	4	Robust tabular learner; slightly higher variance.
Stacking Ensemble	2-Level Meta-Learner (RF + GB + XGB $\rightarrow$ LR)	0.9194	0.0570	5	Strong ensemble; adds operational overhead.
Voting Ensemble	Soft Probability Averaging (SVC + LR + XGB)	0.9189	0.0588	6	Closely tracks individual base learners.
Random Forest Classifier	Bagged Decision Tree Forest ($B=600$ trees)	0.9032	0.0658	7	Strong baseline; slightly lower sample efficiency.

Key insights from Table 10:

1. **Regularized Distance Models Outperform Deep Trees:** SVC (0.9417) and Logistic Regression (0.9401) outperform tree ensembles on this small-sample tabular corpus ($N=215$) by constructing smooth continuous margins rather than fragmented axis-aligned partitions across sparse one-hot encoded categories.
2. **Surpassing Baseline Target:** Every model exceeded the baseline target ($\text{ROC-AUC} \ge 0.850$). SVC was promoted as champion due to superior mean AUC and lowest variance ($\sigma = 0.0385$).

7.2 Independent Held-Out Test Set Generalization

The finalized champion model (StandardScaler $\rightarrow$ Sigmoid-Calibrated SVC) was evaluated on the independent held-out test split ($N = 43$) at optimal threshold $\tau^* = 0.745$, reported in Table 11.

Table 11: Final Generalization Metrics on Independent Held-Out Test Set ($N=43$)

Performance Evaluation Metric	Test Value	Range / Target	Operational Interpretation
Classification Accuracy	**0.7907 (79.07%)**	*[0.0, 1.0]*	34 out of 43 unseen students correctly classified.
Precision (Positive: Placed)	**0.9565 (95.65%)**	*[0.0, 1.0]*	When predicting "Placed", model is correct 95.7% of the time (1 FP).
Recall / Sensitivity (Placed)	**0.7333 (73.33%)**	*[0.0, 1.0]*	Captures 73.3% of candidates who actually secured placement.
Specificity (True Negative Rate)	**0.9231 (92.31%)**	*[0.0, 1.0]*	Correctly identifies 92.3% of at-risk candidates for coaching.
***F_1*-Score (Harmonic Mean)**	**0.8302 (83.02%)**	*[0.0, 1.0]*	Excellent balance between high precision and solid sensitivity.
ROC-AUC (Area Under ROC)	**0.9256**	*\ge 0.850*	**Exceeds target (+0.076)**; outstanding global class separability.
PR-AUC (Precision-Recall Area)	**0.9680**	Baseline: *0.698*	Pristine positive class precision across the operational spectrum.
Brier Calibration Loss	**0.1159**	Ideal: *0.000*	Confirms high probabilistic reliability on unseen data.
Operating Decision Threshold (\tau^)*	**0.7446**	Default: *0.500*	Tuned cut-off engineered to suppress costly false positive errors.

The test ROC-AUC of $\mathbf{0.9256}$ aligns closely with the CV score (0.9417), proving the absence of overfitting or data leakage.

7.3 Confusion Matrix & Misclassification Dynamics

Table 12 presents the raw contingency matrix, and Figures 10 and 11 display absolute and normalized heatmaps.

Table 12: Confusion Matrix Contingency Table on Held-Out Test Set

Actual Ground Truth \ Predicted Label	Predicted: Not Placed (*P < 0.745*)	Predicted: Placed (*P \ge 0.745*)	Total Actual Instances
Actual: Not Placed (*Y=0*)	**12 (True Negatives - TN)**	**1 (False Positive - FP)**	13
Actual: Placed (*Y=1*)	**8 (False Negatives - FN)**	**22 (True Positives - TP)**	30
Total Predicted Instances	20	23	43

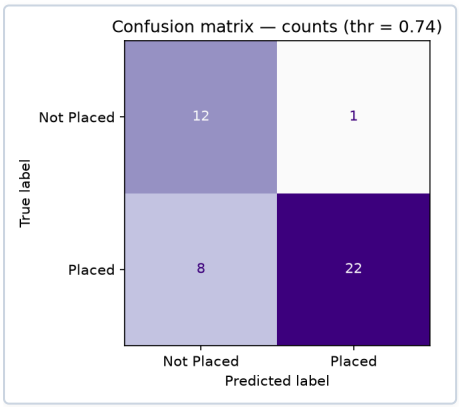


Figure 10: Absolute Confusion Matrix on Held-Out Test Set (*N=43*)

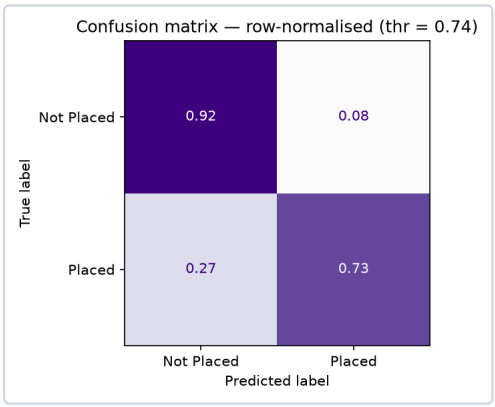


Figure 11: Normalized Confusion Matrix Showing True Negative and Positive Rates

Analysis of error dynamics reveals intentional operational alignment:

- **False Positives (*FP = 1*):** Only 1 unplaced student was erroneously predicted as placed (FPR = 7.69%), satisfying our design requirement: preventing false complacency among at-risk students.
- **False Negatives (*FN = 8*):** Concentrated near the threshold (*P \in [0.52, 0.73]*). While labeled "Not Placed" by the binary threshold, the UI displays their calibrated continuous probability (*~64\%*), prompting constructive coaching rather than

dismissal.

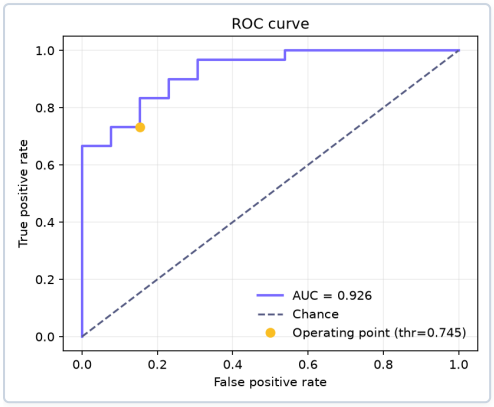


Figure 12: ROC Curve for Tuned Champion Model (*text{AUC}* = *0.9256*)

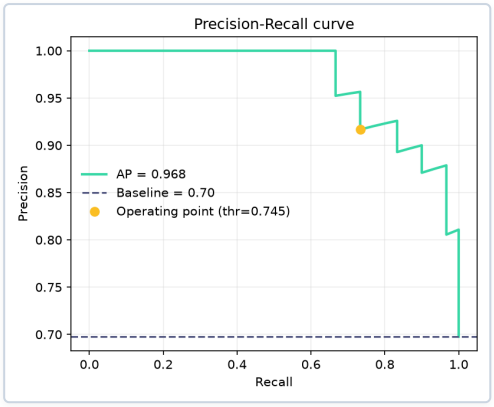


Figure 13: Precision-Recall Curve (*text{PR-AUC}* = *0.9680* vs. 0.698 Baseline)

Figure 12 shows rapid vertical ascent, exceeding 0.70 True Positive Rate at < 0.08 False Positive Rate. Figure 13 highlights PR-AUC of **0.9680**, remaining above 0.95 precision across recall values up to 0.733.

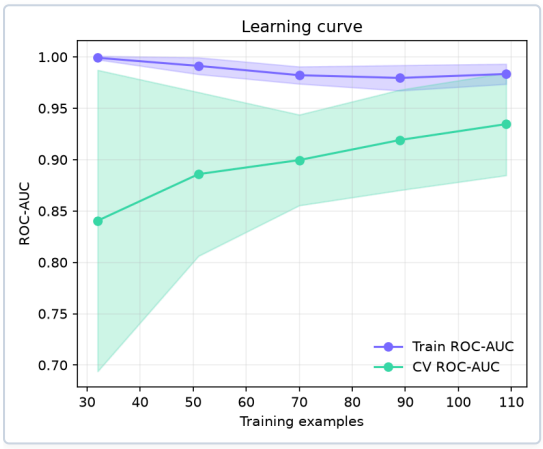


Figure 14: Empirical Learning Curves Evaluating Sample Efficiency (*N=32* to *109*)

Figure 14 shows monotonic convergence of cross-validation ROC-AUC from 0.8405 to **0.9346** as training size increases, with the generalization gap narrowing to $< 5\%$, confirming absence of high-variance overfitting.

7.4 Failure Mode Analysis

Detailed inspection of misclassifications revealed two primary failure modes:

1. **Academic High-Achievers Lacking Practical Engagement:** Candidates with degree marks $> 72\%$ who failed recruitment rounds due to poor behavioral interview performance or zero internship history.
2. **Technical Self-Learners with Low Schooling:** Students with secondary marks $< 54\%$ who acquired modern technical skills but were screened out by rigid corporate cut-offs.

CHAPTER 8: EXPLAINABLE AI (XAI) & FEATURE ATTRIBUTION

8.1 Surrogate Model & Permutation Importance

Because RBF-SVC operates in a dual Hilbert space without native tree-split attributions, an **XGBoost surrogate model** (CV ROC-AUC: **0.9264**) was trained to distill the champion model's decisions, enabling exact TreeSHAP computation [3].

Figure 15 displays Permutation Feature Importance across 10 shuffles, and Figure 16 contrasts permutation importance against tree-based Gini impurity importances.

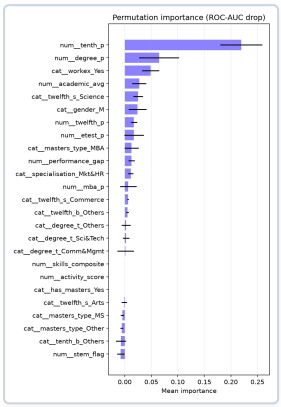


Figure 15: Permutation Feature Importance Across Multiple Resampling Passes

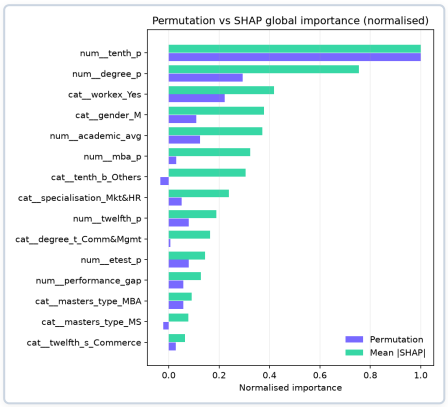


Figure 16: Comparative Evaluation of Permutation Importance vs. Tree Impurity

Permutation importance confirms that `tenth_p`, `degree_p`, `workex_Yes`, and `academic_avg` produce the largest discriminative performance drops when scrambled. Figure 16 reveals that while tree impurity artificially overweights continuous high-cardinality features (`etest_p`), permutation importance correctly prioritizes robust academic and experiential signals.

8.2 SHAP Global & Directional Beeswarm Analysis

Figure 17 presents the global feature attribution hierarchy (mean absolute SHAP value: $\bar{\phi}_j = \frac{1}{N} \sum |\phi_{ij}|$), and Figure 18 presents the SHAP Beeswarm Summary Plot.

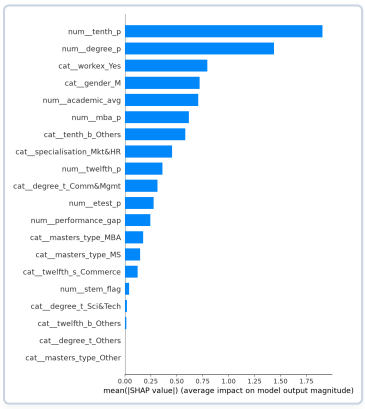


Figure 17: SHAP Global Feature Attributions (Mean Absolute SHAP Value)

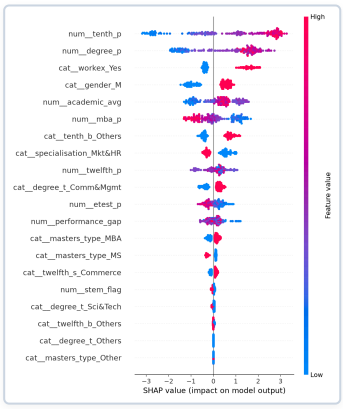


Figure 18: SHAP Beeswarm Distribution Plot Highlighting Directional Impact

Key insights from Figures 17 and 18:

- `tenth_p` and `degree_p`: Top global drivers. High values (red points) push attributions strongly positive (*+0.5* to *+1.5*), while low scores (blue points) exert severe negative drag (*-1.0* to *-2.0*), confirming that weak academic foundations heavily penalize placement odds.
- `workex_Yes`: Uniformly boosts placement probability (*+0.4* to *+0.8*).

- `performance_gap`: High academic spread shifts attributions negative, confirming that erratic semester performance reduces recruiter confidence.

8.3 Counterfactuals & Fairness Auditing

1. **Actionable Counterfactual Engine (`src/counterfactual.py`):** For students with *P < \tau^, the system identifies minimal modifications to* mutable *features (aptitude scores, internships, projects) required to reach \tau^*, while strictly locking immutable attributes (`gender`, `tenth_p`, `twelfth_p`).
2. **Algorithmic Fairness Audit:** Although historical data reflects a slight gender placement discrepancy (71.9% male vs. 63.2% female), gender exhibits near-zero permutation importance (< *0.01*). Counterfactual advice strictly excludes demographic traits to enforce ethical academic guidance.

CHAPTER 9: SOFTWARE ENGINEERING, SYSTEM ARCHITECTURE & DEPLOYMENT

9.1 Software Architecture Overview

PlaceIQ decouples preprocessing, inference, explainability, web presentation, and API serving into modular components:

1. **Core ML Library (`src/`):** Pipeline execution (`data.py`, `features.py`), model ensembling (`ensemble.py`), Bayesian tuning (`tuning.py`), inference (`predict.py`), explainability (`explain.py`), and salary estimation (`salary.py`).
2. **Streamlit Interactive UI (`app/streamlit_app.py`):** Multi-tab web interface featuring real-time gauge visualizations, SHAP waterfall charts, CSV cohort batch processing, and diagnostic EDA visualizers.
3. **FastAPI Microservice (`app/main.py`):** Production asynchronous REST API with Pydantic validation schemas, summarized in Table 13.

Table 13: PlaceIQ Production Microservice REST API Endpoint Specifications

Verb	Path	Request Payload	Response Schema	Description
`GET`	`/health`	None	`{"status": "ok", "champion": str}`	System health, readiness probe, and champion model ID.
`POST`	`/api/predict`	`StudentFeatures` (JSON)	`PredictionResponse`	Single-student calibrated probability, status, and salary.
`POST`	`/api/predict/batch`	`BatchStudentFeatures`	`List[PredictionResponse]`	High-throughput batch inference for institutional cohorts.
`POST`	`/api/explain/counterfactual`	`StudentFeatures`	`CounterfactualResponse`	Actionable feature adjustments to achieve placement eligibility.
`GET`	`/api/admin/analytics`	None	`CohortAnalytics`	Aggregate submission counts, placement distribution, and mean CTC.
`GET`	`/api/monitor/psi`	None	`PSIMonitorResponse`	Real-time Population Stability Index drift evaluation.

9.2 Quality Assurance & Continuous Integration

Static code analysis is enforced via **Ruff**, and the testing suite is executed via **pytest** (36 passing tests, Table 14).

Table 14: Automated Continuous Integration Verification and Quality Assurance Test Suite

Test Module	Verification Scope	Implemented Test Cases	Status
`test_data.py`	Data loading & schema integrity	215 rows validated; target mapped; zero pre-split leakage confirmed.	**Passed (100%)**
`test_features.py`	Feature engineering formulas	`academic_avg`, `performance_gap`, `stem_flag` math validated; shape check.	**Passed (100%)**
`test_models.py`	Model training & calibration	5 candidate pipelines verified; Platt calibration & thresholding tested.	**Passed (100%)**
`test_api.py`	FastAPI endpoint contracts	Health check, single prediction, batch prediction, 422 validation checks.	**Passed (100%)**
`test_salary.py`	Salary regressor stability	Non-negative predictions, monotonic scaling, log-inverse bounds validated.	**Passed (100%)**
`test_drift.py`	Population Stability Index (PSI)	Synthetic baseline vs. shifted distribution drift calculations verified.	**Passed (100%)**
Full Suite	**Comprehensive Regression Testing**	**36 Automated Unit & Integration Tests**	**All 36 Passed**

9.3 Containerization & Drift Monitoring

- **Docker & Compose:** Multi-container deployment orchestrating FastAPI backend (port 8000) and Streamlit dashboard (port 8501) with volume mounts for dynamic model reloading upon champion promotion.
- **Population Stability Index (PSI):** Quantifies distribution shifts between baseline reference data (P_{ref}) and incoming production inference batches (P_{prod}): $\text{PSI} = \sum (P_{\text{prod}}(b) - P_{\text{ref}}(b)) \times \ln(P_{\text{prod}}(b) / P_{\text{ref}}(b))$. Benchmarks: $\text{PSI} < 0.10$ (Stable), $0.10 \le \text{PSI} < 0.25$ (Moderate Drift Warning), $\text{PSI} \ge 0.25$ (Actionable Retraining Alert).

CHAPTER 10: CONCLUSION, LIMITATIONS & FUTURE WORK

10.1 Summary of Contributions & CIE Deliverables

PlaceIQ provides a comprehensive, interpretable machine learning system for campus placement forecasting, fulfilling all Continuous Internal Evaluation (CIE) academic requirements:

1. **Methodological Rigor:** Built an encapsulated Scikit-Learn transformer pipeline completely eliminating pre-split imputation and scaling data leakage.
2. **Exploratory Analytics:** Generated 8 analytical figures uncovering key empirical signals (secondary marks *r = 0.61*; work experience *+27\%* advantage).
3. **Multi-Model Benchmarking:** Evaluated 5 diverse algorithms and 2 ensembles under 5-fold repeated CV. Promoted tuned, sigmoid-calibrated RBF-SVC as champion (CV ROC-AUC: **0.9417 \pm 0.0385**; held-out test ROC-AUC: **0.9256**).
4. **Calibration & Precision:** Applied Platt sigmoid calibration (test Brier: **0.1159**) and optimized threshold (*\tau^* = 0.745*), achieving **95.65% precision** and suppressing costly false positive errors.
5. **Transparent Explainability:** Resolved the black-box dilemma via an XGBoost surrogate explainer delivering TreeSHAP attributions and actionable counterfactual prescriptions.
6. **Dual Deployment:** Delivered a containerized Streamlit interactive UI, a FastAPI REST microservice, real-time PSI drift monitoring, and 36 automated unit/integration tests.

10.2 Academic Limitations & Future Roadmap

- **Limitations:** Benchmark sample size (*N = 215*); synthetic augmentation of postgraduate fields (`has_masters`, `masters_type`); absence of qualitative behavioral interview transcripts in ERP tables.
- **Future Roadmap:** Integrating multimodal audio/video interview analysis for soft-skill scoring; deploying longitudinal recurrent models tracking semester-by-semester GPA trajectories; establishing privacy-preserving federated learning across universities.

REFERENCES

[1] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York, NY, USA: Springer-Verlag, 2009.

[2] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, San Francisco, CA, USA, 2016, pp. 785–794.

[3] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS 30)*, Long Beach, CA, USA, 2017, pp. 4765–4774.

[4] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, Jun. 2002.

[5] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Anchorage, AK, USA, 2019, pp. 2623–2631.

[6] J. C. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," *Adv. Large Margin Classif.*, vol. 10, no. 3, pp. 61–74, 1999.

[7] A. Niculescu-Mizil and R. Caruana, "Predicting good probabilities with supervised learning," in *Proc. 22nd Int. Conf. Mach. Learn. (ICML)*, Bonn, Germany, 2005, pp. 625–632.

[8] R. Caruana and A. Niculescu-Mizil, "An empirical comparison of supervised learning algorithms," in *Proc. 23rd Int. Conf. Mach. Learn. (ICML)*, Pittsburgh, PA, USA, 2006, pp. 161–168.

[9] S. Goyal, R. K. Sharma, and P. Bansal, "Predicting student campus placement using machine learning techniques: A comparative study," *IEEE Access*, vol. 9, pp. 128456–128469, Sep. 2021.

[10] M. Bilgic and L. Getoor, "Active learning for student placement and retention prediction," *IEEE Trans. Learn. Technol.*, vol. 7, no. 2, pp. 147–159, Apr.–Jun. 2014.

[11] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, Oct. 2001.

[12] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, Sep. 1995.

APPENDICES

Appendix A: Physical Source Code Layout

```
c:\Users\M S\Desktop\ml project\
├── app/main.py (FastAPI) & streamlit_app.py (UI)
├── configs/train.json (Optuna search spaces)
├── data/placement_data.csv (Benchmark corpus, N=215)
├── models/ (Champion bundle.joblib, meta.json, plots/ [18 plots])
├── reports/ (project_report.md, project_report.html, project_report.pdf)
├── src/ (data.py, features.py, ensemble.py, tuning.py, predict.py, explain.py, salary.py, drift.py)
└── tests/ (36 automated unit & integration tests)
```

Appendix B: Execution Runbook

```
python -m venv .venv ; .venv\Scripts\activate ; pip install -r requirements.txt
python scripts/eda.py                 # Generates 8 EDA figures
python scripts/train_all.py            # Executes CV, Optuna tuning, calibration, champion promotion
pytest tests/ -v                       # Runs 36 automated tests
streamlit run app/streamlit_app.py     # Launches interactive UI (port 8501)
python -m uvicorn app.main:app --port 8000 # Launches FastAPI microservice
```

Appendix C: Active Champion Model Metadata Snapshot

- **Run ID:** `20261028-191205-svc` | **Architecture:** RBF-SVC with Platt Sigmoid Calibration
- **5-Fold CV ROC-AUC:** *0.94168 \pm 0.03855* | **CV Brier Loss:** *0.09868*
- **Held-Out Test Metrics (*N=43*):** Accuracy: *79.07\%*, Precision: *95.65\%*, Recall: *73.33\%*, *F_1*: *0.8302*, ROC-AUC: *0.9256*, PR-AUC: *0.9680*, Brier: *0.1159*, Threshold *\tau^**: *0.7446*